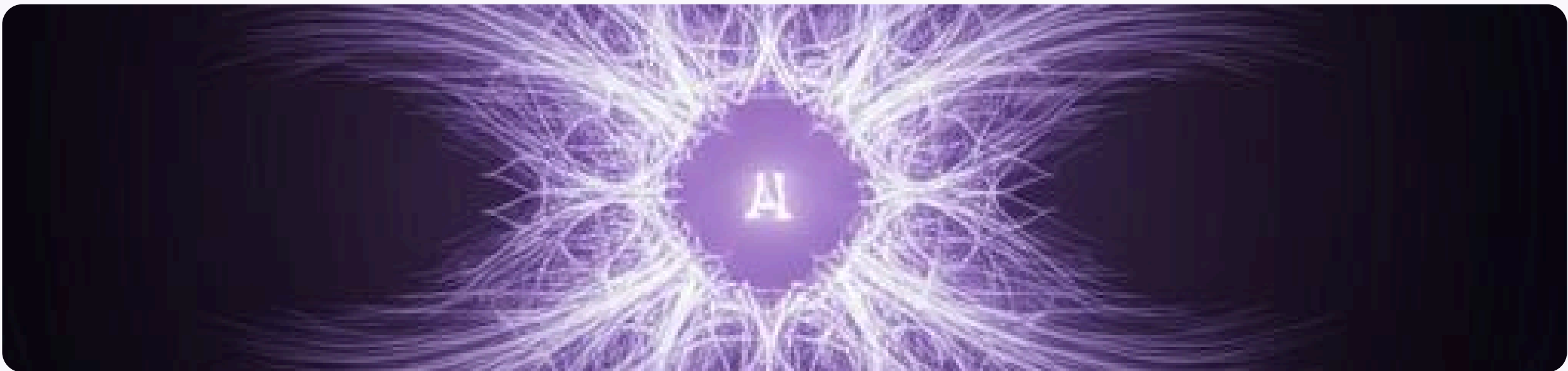


Bag of Visual Words for Image Classification

A Computer Vision Project using SIFT, KMeans, KNN & SVM



Overview

This project explores image classification using the Bag of Visual Words (BoVW) approach. It leverages Scale-Invariant Feature Transform (SIFT) for feature extraction and KMeans clustering to create a visual vocabulary. The system classifies images of cats, chairs, and cameras.

Objective

- ☐ Extract distinctive features from images using SIFT.
- ☐ Cluster features to form a visual vocabulary.
- ☐ Train classifiers using K-Nearest Neighbors (KNN) and Support Vector Machines (SVM).
- ☐ Classify images into categories: cat, chair, and camera.

Tools and Technologies

 Python

 OpenCV

 SIFT

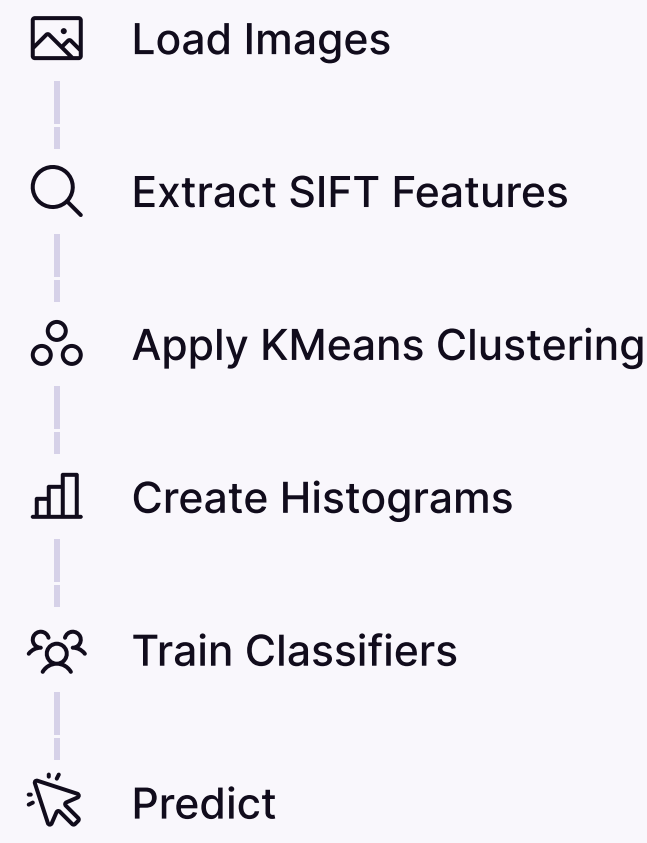
 KMeans

 KNN

 SVM

 Matplotlib

Methodology Flowchart



Visual Vocabulary (KMeans)

KMeans clustering is applied to the extracted SIFT descriptors to create a visual vocabulary. Each cluster represents a visual word, and the frequency of these words in an image forms its feature vector.

Classification Results

Classifier	Cat	Chair	Camera
KNN	85%	78%	92%
SVM	90%	82%	95%

Visualizations



Predicted Class Labels for Test Images (Cat, Chair, Camera)

Key Learnings

This project taught me how to represent images using bag-of-words techniques, and how clustering + classical ML can be combined to build powerful vision systems.